Phil McDonough
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Data scientist with an MS in Analytics and a focus on applied AI research. Experienced in taking AI problems from product requirements to prototype models, including data preparation and annotation, experiments with LLMs and fine-tuned encoders, and the evaluation and feedback loops that drive improvement. Additional experience in NLP on large-scale text and responsible-AI testing.

Skills: Python, R, SQL
Certifications: PCEP—Certified Entry-Level Python Programmer, AWS Certified Cloud Practitioner

EXPERIENCE

Axon Remote
AI Research Contractor June 2025–Present

- Took an early-stage AI product from requirements to a working prototype, building the benchmark dataset, annotation guidelines, and reusable evaluation infrastructure used to measure progress against product-relevant metrics.
- Conducted LLM model and prompt ablations and failure-pattern analysis to evaluate tradeoffs against product-relevant metrics. Developed synthetic data generators and fine-tuned a BERT-based encoder that performed comparably to the LLM-based prototype.
- Built annotation tooling and a feedback loop to analyze model predictions and generate targeted training data, supporting continuous model improvement.
- Designed counterfactual dialect evaluations to identify performance disparities and tested targeted retraining to address them. Adapted a PII redaction approach, added a replacement component, and evaluated the full workflow.
- Benchmarked voice activity detection models, tested model conversion and on-device feasibility, and studied how speech filtering affected downstream alerts.

PROJECTS

Fifth Third Bank Practicum Raleigh, NC
Tech Lead August 2024–April 2025

- Analyzed an unstructured text data set with 7 million bankers notes, exploring both LLM and NLP approaches.
- Created features using NLP techniques such as LDA to cluster notes based on customer relationship health and product interest.
- Integrated features with pre-existing customer data to predict product adoption using boosting algorithms.

Elevance Health Capstone Blacksburg, VA
Team Member August 2022–December 2022

- Collaborated with the Individual Strategy and Analytics team to increase their understanding of market sizes in the US for the five main insurance types at the county level using Python for data collection and processing, and Tableau for dashboard visualizations.
- Executed a comprehensive 'Tools and Techniques' presentation, highlighting the array of aforementioned tools and techniques employed by our project team.

EDUCATION

MS in Analytics May 2025
Institute for Advanced Analytics, NC State University, Raleigh, NC

BS in Computational Modeling and Data Analytics, Economics Focus, Mathematics Minor May 2023
magna cum laude
Virginia Tech, Blacksburg, VA